

PAAMIUT FREDERIKSHAAB, Greenland

WMO: 042600

Lat: 62.015N

Lon: 49.671W

Elev: 49

StdP: 14.67

Time Zone: -3.00 (GRL)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.1	2.9	-16.2	2.3	3.5	-12.1	2.9	7.3	40.1	24.2	35.9	24.7	6.6	20	0.867

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	56.4	47.5	53.4	46.5	51.0	45.4	49.4	53.2	47.8	51.5	46.4	49.8	11.6	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
47.1	48.0	50.4	45.5	45.1	48.8	44.0	42.6	47.5	20.0	53.4	19.1	51.5	18.4	50.1	61.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.0	28.5	24.6	DB	-3.4	63.1	7.2	4.8	-8.5	66.5	-12.7	69.4	-16.8	72.1	-22.0	75.6	(4)
(5)			WB	-5.0	52.7	6.6	2.8	-9.8	54.7	-13.6	56.3	-17.3	57.9	-22.1	59.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	31.9	21.3	20.0	21.6	29.7	36.0	40.8	43.8	43.7	40.1	34.1	27.5	23.5
	DBStd	11.09	9.00	9.95	9.58	6.87	5.16	4.16	4.03	4.04	4.05	5.76	7.57	8.49
	HDD50	6623	891	841	880	609	437	277	198	199	298	495	676	822
	HDD65	12082	1356	1261	1345	1059	900	725	657	660	747	959	1126	1287
	CDD50	14	0	0	0	0	1	1	6	4	1	1	0	0
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	9.7	10.6	11.4	10.9	10.8	9.6	8.1	7.4	7.8	9.8	9.4	9.9	10.2
Precipitation	PrecAvg	52.5	5.4	5.3	4.2	3.6	3.1	3.0	3.0	3.9	5.2	4.5	5.9	5.3
	PrecMax	88.1	14.4	13.6	11.4	9.3	7.8	5.9	7.2	8.8	10.3	10.4	12.9	13.8
	PrecMin	29.2	1.5	1.4	0.6	0.7	1.0	0.2	0.3	1.1	1.5	1.6	1.3	1.3
	PrecStd	11.4	3.1	3.2	2.4	1.9	1.4	1.6	1.7	1.8	2.1	2.3	3.2	2.8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	47.7	48.4	48.2	50.6	57.5	56.8	60.3	59.2	57.9	55.5	51.2	47.1
		MCWB	38.9	42.2	40.0	39.2	46.1	46.6	51.1	50.7	46.4	44.8	42.3	37.8
	2%	DB	42.1	42.8	43.1	45.5	50.2	52.5	55.3	54.6	51.7	48.3	44.5	42.0
		MCWB	34.9	35.7	36.7	37.7	43.2	44.9	48.0	48.3	44.5	40.3	37.9	35.4
	5%	DB	37.0	37.9	38.7	42.0	46.0	49.5	52.5	51.9	48.1	44.4	40.0	38.3
		MCWB	32.0	32.2	33.4	36.8	40.6	44.2	46.7	47.1	43.9	39.0	35.1	33.8
	10%	DB	33.8	34.0	35.0	38.9	43.2	47.2	50.0	49.7	45.8	41.7	36.9	35.3
		MCWB	29.8	30.2	31.5	34.8	39.1	43.0	45.8	46.0	42.1	37.7	33.5	31.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	40.2	42.1	42.0	42.3	47.9	48.5	52.2	51.8	48.7	45.5	43.6	40.3
		MCDB	45.9	48.1	46.5	46.3	53.7	53.7	57.5	57.4	53.2	52.0	50.2	44.1
	2%	WB	35.7	36.8	37.5	39.6	43.8	46.4	49.2	49.4	46.3	41.9	39.4	36.7
		MCDB	40.7	40.5	41.8	43.4	49.4	50.3	53.1	52.8	50.2	45.4	44.3	40.4
	5%	WB	32.9	33.4	34.0	36.9	41.4	44.7	47.4	48.0	44.2	39.7	35.9	34.1
		MCDB	36.7	37.3	37.9	40.5	45.2	48.6	51.2	51.1	47.5	43.0	38.3	37.7
	10%	WB	30.0	30.3	31.2	35.1	39.5	43.3	46.0	46.5	42.6	38.0	33.8	31.9
		MCDB	33.2	33.5	34.8	38.5	42.5	46.8	49.5	49.1	45.3	41.1	36.5	34.9
Mean Daily Temperature Range	5% DB	MDBR	9.5	10.2	10.2	8.8	7.3	8.0	8.0	7.3	7.1	8.0	8.9	9.1
		MCDBR	14.0	14.5	15.5	14.3	13.3	13.0	13.2	11.6	12.0	12.3	12.9	11.8
		MCWBR	9.9	10.1	10.7	9.1	7.6	7.3	7.6	6.7	7.1	7.6	9.4	7.9
	5% WB	MCDBR	13.8	14.4	15.1	12.7	11.7	11.7	11.7	10.5	10.7	10.8	11.9	11.2
		MCWBR	10.4	11.1	11.1	9.0	7.6	7.4	7.5	6.9	7.0	7.7	9.5	8.3
Clear-Sky Solar Irradiance	taub	0.235	0.272	0.292	0.334	0.349	0.352	0.360	0.357	0.322	0.284	0.227	0.366	
	taud	2.027	2.249	2.326	2.261	2.325	2.391	2.385	2.418	2.451	2.365	2.092	2.794	
	Ebn at Noon	139	223	262	270	275	277	271	262	251	216	143	102	
	Edh at Noon	12	19	26	34	35	34	33	29	23	17	10	7	
All-Sky Solar Radiation	RadAvg	74	289	764	1363	1796	1953	1808	1355	839	434	124	39	
	RadStd	7	30	64	107	90	118	142	104	78	33	16	3	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.10	+5.22	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)		+1.10	+5.22	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page